

Backup Software

Backup software have just as important a role to play as does the hardware. All the best and fastest hardware in the world will not help much if you choose a software that does its job inadequately. Though companies with really big backup requirements could have existing software customised to their setup, there are a few good software that will do the job straight out of the box.

Most hardware vendors will provide you with their own software when you purchase their hardware, and these are often good enough. You should look for distributed backup software as a total backup software solution for your company.

Distributed backup software will give you, at the head office, total control over backup policies at your branch offices. This is done via the Internet, and lets the head office engineers dictate a uniform backup policy for all offices. System engineers at branch offices have only the task of managing the servers and media (tape, optical media, external drives, etc.), and other routine local tasks.

Computer Associates (www.cai.com), Veritas (www.veritas.com), Legato (www.legato.com) and HP (<http://welcome.hp.com/country/in/en/welcome.html>) are just a few of the biggest names that offer easy-to-set up and robust enterprise-level backup software solutions.

NAS (Network Attached Storage)

Basically, a NAS is nothing more than a very simple computer with many hard disks and a simple OS, used mainly to increase the capacity of existing servers, or to supplement the storage capacity for users over a LAN. The hardware and OS are made specifically to enable fast file transfers and handle many simultaneous requests. A NAS device can also be used as a backup device that can be plugged in when backups are needed, and stored away for safety after the backups are done. The beauty of a NAS is that it costs less than a backup server, while still providing all the desired functions.

NAS devices are available from all the top vendors in the country, such as HP and Iomega. A typical NAS generally offers anywhere from 250 GB to 2 TB of storage space. The advantage of a NAS is that it uses drive-based storage, which offers data transfer speeds that are much higher than traditional tape-based storage. Also, NAS devices are very easy to add to an existing LAN, as they are more or less plug-and-play.

Tape Drive Servers

These servers use magnetic tapes to store large volumes of data. They are great for high volumes of data storage, but have slower read and write speeds. Tape media is also a high-risk storage medium, as the media is frail, and needs to be stored carefully.

Systems engineers also need to constantly monitor the media and avoid overuse of a tape, as this could lead to wearing away of the magnetic layer and thus, loss of data. Also, tape drives need to have regular maintenance done, especially the heads that read the tape—these need frequent replacement or cleaning.

The biggest advantage of tape servers has always been cost. For volumes of data backup in terms of terabytes, nothing beats tape on per-GB cost. Also, the portability of tape cartridges means that off-site storage of archived data is easier, faster and cheaper.

Hard Drive Servers

Hard drive servers have fast read and write speeds, and require less maintenance than tape drives. However, the drawback here is cost, lack of portability, and the maximum capacities. Most companies use them as file servers for data that requires constant access, due to the higher speeds they offer. They aren't as cost-effective as tape drives, but are a lot more reliable. As far as portability goes, even with a Hot Swap drive server, the safest place to store the hard disks is probably inside the server cabinet itself!

For a company that needs to have a short turnaround time after a disaster, disk drives are often the only alternative. As businesses turn global, data is required 24/7, and not all companies can afford the time spent on backing up to or recovering from tape servers. For such scenarios, though more expensive, disk servers offer the required data access speed, both while backing up and while recovering lost data.

What Do I Need?

Now comes the hardest part—choosing a solution, or a combination of solutions, for your company. There are three broad categories that any business would fit into, based upon the amount of data that needs to be backed up, and the frequency at which backups are performed. Let's take a look at the three hypothetical companies and their backup solution setups.

Small Business

Company A has two or three network servers, and has less than 80 GB of critical data. Their daily backup is less than 40 GB, and they require archiving of data at least once a month. The company has a per-user backup policy—each user is solely responsible for backing up his or her own data on the servers. The solution they use is an additional network server with dual hard drives running with RAID 1 configuration. Backups are taken at the end of every working day. At the end of the month, data for archiving is manually burnt onto DVD media.

Company A runs personal software on every user's computer to back up data onto their network file servers. Then, software on the backup server copies the data from the network servers—it is scheduled to do so about an hour



Online Storage

With broadband penetration in India increasing, online backup solutions are looking more attractive, especially for offices. The advantage of this solution is that your data is accessible from anywhere in the world—only to you. It is also the most convenient in terms of ease of use and infrastructure. All you need to do is run the provided software, select which folders you want backed up, and then back up the data. There is no hassle about maintaining hardware. The drawbacks are that you are dependant solely on your Internet connection, and you might not trust the Internet for critical company data that may contain trade secrets, proposals, etc.

The costs are falling here too, but are still a lot more than using tape or hard drives as backup solutions. If you do opt for this backup technique, make sure to check for hidden costs, data redundancy at the provider's end, encryption and security, and, of course, company reviews and experiences.

Fifty GB of backup space can cost anywhere between Rs 80,000 and Rs 3,80,000, depending on whether you need a single-user or workgroup backup solution. Some online backup providers are: www.ibackup.com or www.xdrive.com.